

Bayesian Estimation of a Dynamic Model of Two-Sided Markets: Application to the U.S. Video Game Industry

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Abstract. This paper develops and estimates a structural model of two-sided markets with durable platform intermediaries and affiliated products. It models buyers' purchase decisions of platforms and affiliated products and sellers' decisions of price setting and entry, accounting for the dynamic interaction between the two distinct groups of platform participants. To estimate the proposed model, this paper develops a Bayesian Markov chain Monte Carlo estimation approach that incorporates nonparametric approximation and interpolation methods. The proposed model and estimation method are applied to the 32/64-bit generation of the U.S. video game industry. The results of counterfactual experiments show that the dynamic behavior of platform participants has significant impacts on platform adoption and the affiliated product market, and that a failed platform could have survived if it had priced the two sides properly in a dynamic two-sided market environment.

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1. Introduction

In many two-sided or “platform” markets, platforms serve two distinct categories of customers, called buyers and sellers: they can interact only if they have joined the same platform. Buyers adopt a platform to access products provided by sellers who have joined or will join that platform, and sellers join a platform to reach buyers who have adopted or will adopt that platform. Buyers' valuation of a platform depends on the availability, quality, and prices of that platform's affiliated products. The success of a platform's affiliated products depends on that platform's installed base of buyers. In literature on two-sided markets, this interdependence, or externality between two distinct groups of participants that a platform serves, is called indirect network effects. Moreover, platform markets are often inherently dynamic environments due to the durability of platform intermediaries and affiliated products. Two-sidedness and dynamics are important features of many markets such as operation systems and software, DVD players and DVDs, video game consoles and games, e-readers and e-books, smartphones and apps, and many others. This paper models the dynamic strategic interaction between buyers and sellers in these types of two-sided markets, develops a novel method to estimate the proposed model, and applies it to the U.S. video game industry.

This paper emphasizes the strategic interdependence between the two groups of platform participants. On the buyer side, buyers make platform adoption decisions and, if they adopt a platform, the purchase decisions of the affiliated products. The value of adopting a platform depends on the value of being able to purchase the products affiliated with that platform, so buyers rationally anticipate the affiliated product market when they make their platform adoption decisions. On the seller side, potential entrants decide whether to enter, and incumbent sellers decide how much to charge. Only those buyers who have adopted a platform are potential customers of the affiliated products, so sellers' entry and price choices depend on buyers' platform adoption.

This paper also stresses the need to model the platform participants as forward-looking agents, mainly because it focuses on those two-sided markets with durable platform intermediaries and affiliated products. Almost all theoretical and empirical studies in the literature of durable goods have modeled buyers and sellers as forward-looking agents.¹ Buyers have heterogeneous valuations of a durable good. They make their purchases at different points in time, and, as a result, the composition of remaining buyers changes over time. This gives sellers an incentive to engage in intertemporal price discrimination, that is, setting high initial prices to sell to buyers with high valuations

and then cutting prices to appeal to buyers with low valuations. A static pricing model cannot explain the declining pattern of prices that has been documented in the current and many previous studies on durable goods, including Gandal et al. (2000) on CD players, Nair (2007) and Lee (2013) on video games, and Conlon (2012) on LCD TVs. Moreover, when prices are declining over time, buyers have an option value associated with waiting. Estimating a static demand model for durable goods would give biased price elasticities and hence biased predictions.

Failure to account for platform participants' dynamic behavior in two-sided markets would generate additional biases. On the buyer side, when buyers make platform adoption decisions, they anticipate not only the utility they will derive from the affiliated product that is currently available, but also that from products that will be released in the future. Ignoring this dynamic behavior would misspecify the platform values that depend on the affiliated product market, and hence would give upward-biased estimates for qualities of products released early and downward-biased estimates for qualities of products released late. On the seller side, when sellers make entry decisions, they trade off between the entry cost and the discounted profit that depends not only on a platform's current installed base of buyers but also on buyers' platform adoption in the future. Estimating a static model of dynamic entry would give downward-biased estimates of entry costs.

To estimate the proposed model, this paper develops a new practical method for structural estimation of dynamic models. In an equilibrium of a dynamic model, given other agents' strategies, each agent's best response is the solution to a single-agent dynamic programming problem. Moreover, an equilibrium is a fixed point of the system of best response operators. For relatively complicated models, calculating the continuation value and policy function is computationally difficult, or even impossible. The proposed estimation method combines the Bayesian algorithm and the dynamic model solution algorithm into a single algorithm that estimates the parameters and solves the model simultaneously. The idea was originally proposed by Imai et al. (2009; hereafter, IJC), who introduced a Bayesian method to the estimation of discrete choice dynamic programming models. In the outer loop, a Metropolis–Hastings algorithm is used to draw a sequence of parameter vectors from their posterior distributions. In the inner loop, for a given parameter vector along the Markov chain Monte Carlo (MCMC) chain, each agent's value function and perceived policy function are approximated using the value functions and policy functions from previous MCMC iterations, respectively. Given other agents playing their perceived equilibrium strategies, each agent's value function and policy function are updated and stored for

future MCMC iterations. An interpolation approach similar to that of Keane and Wolpin (1994) is adopted to cope with the state space problem. This estimation procedure does not fully solve the dynamic model but incorporates the approximation and the interpolation methods. This idea is similar to that used by Pakes and McGuire (2001), who approximated the continuation value using the average of the returns from past outcomes of the algorithm, and updated the value and policy functions at a recurrent class of points, rather than at all possible points, in the state space. The estimation method presented in this paper can be applied to estimating other dynamic models.

The model framework and estimation method are applied to a canonical two-sided market with a durable platform and affiliated products—the 32/64-bit generation of the U.S. video game industry (1995–2000). In the video game market, consoles (e.g., Sony PlayStation (PS)) act as platforms that interact with two groups of participants: buyers and video game sellers (e.g., EA Sports). The 32/64-bit generation was dominated by three console platforms: Sega Saturn, Sony PlayStation, and Nintendo 64 (N64). Counterfactual experiments are conducted to explore the implications of platform participants' dynamic behavior on platform adoption and the affiliated product market, and to explore what might have contributed to Sega's failure even though it was the first mover and an incumbent with a good reputation.

This paper contributes to the literature on Bayesian estimation methods that have been commonly applied to static discrete choice models with latent variables (e.g., Albert and Chib 1993, McCulloch and Rossi 1994, Romeo 2007, Jiang et al. 2009). Imai et al. (2009) and Norets (2009) pioneered the use of Bayesian estimation methods for dynamic discrete choice models. They both used previous outcomes to approximate the value function at current iterations. This paper extends their idea to the estimation of models with dynamic strategic interactions between agents. The estimation procedure does not fully solve the model, but incorporates the approximation approach. Hence, it significantly alleviates the computational burden and makes the Bayesian estimation method applicable to complicated models.

This paper also belongs to the literature on two-sided markets that has been growing quickly in the last decade. Rysman (2009) provides a review of this literature. Previous theoretical work includes Rochet and Tirole (2006), Caillaud and Jullien (2003), Armstrong (2006), Hagiu (2006), Weyl (2010), and many others. More and more empirical papers have emerged recently, studying various industries, including the market of CD players and CD titles (Gandal et al. 2000), the personal digital assistant and compatible software systems (Nair et al. 2004), the yellow

page market (Rysman 2004), electronic payment systems (Gowrisankaran and Stavins 2004, Akerberg and Gowrisankaran 2006), and the health insurance market (Ho 2009).

In particular, this paper adds to the empirical literature on the video game industry. Nair (2007) estimated a dynamic demand model using hundreds of PlayStation games and analyzed video game sellers' intertemporal price discrimination behavior. Ishihara and Ching (2012) quantitatively measured the impact of used games on the new game market. Lee (2013) studied the impact of vertically integrated and exclusive software. Derdenger (2014) studied the impact of providing in-house games on the price competition among console makers. Clements and Ohashi (2005) and Dube et al. (2010) estimated two equations simultaneously: one for console adoption and the other for game variety. In this strand of literature, this paper is closest to Liu (2010), which studies the dynamic pricing competition between two major console makers, Sony and Nintendo. In contrast, this paper focuses on the dynamic interaction between the two sides of the market and hence models them as forward-looking agents. Different from most previous studies that measure the value of affiliated products in a platform's utility function by the number of affiliated products, this paper constructs a measurement that is internally consistent with game sellers' price and entry choices—the total utility from being able to purchase its affiliated products that have been released or will be released.

The rest of this paper is organized as follows. Section 2 presents a general model of two-sided markets that captures the dynamic strategic interactions between two distinct groups of participants. Section 3 describes the Bayesian MCMC estimation procedure and discusses the related computational issues. Section 4 applies the model framework and estimation method to the U.S. video game industry. Section 5 concludes.

2. A Model of Two-Sided Markets

Consider a two-sided market in which both platform intermediaries and affiliated products are durable goods. As stressed in the introduction, all platform participants are forward looking: specially, buyers "time" their platform adoption and affiliated product purchases, incumbent sellers dynamically choose their prices, and new sellers make their entry decisions anticipating future profits. To make the model tractable, I focus on the dynamic strategic interactions between buyers and sellers, rather than developing a full dynamic equilibrium model that requires solving for platforms' optimal choices by taking into account buyers' dynamic adoption decisions and sellers' dynamic entry and pricing decisions. Therefore, I model only one generation of platforms, for example,

five or six years for video game consoles and one or two years for e-readers, smartphones, and operation systems. Following Hagiu (2006) and Lee (2013), I assume that a finite number of platform providers, indexed by $l = 1, \dots, L$, preannounce and commit to their choices including their entry time, entry fees to each side, and transaction fees. After the actual release of platforms, the following events occur in each time period.

(i) Each incumbent seller chooses a price. Each potential entrant draws an entry cost from a known distribution and decides whether to enter. If it enters, it will earn profit starting from the next period. Price and entry decisions are made simultaneously.

(ii) Buyers immediately observe the prices but not the entry outcomes. Buyers who have not adopted any platform decide whether to pay an entry fee to adopt one. Each buyer is allowed to adopt at most one platform within a generation.² Once she has adopted a platform, she becomes a potential buyer for the affiliated products.

(iii) Sellers' entry decisions are implemented. We move to the next period.

The affiliated product market consists of a finite number of separate submarkets. Each buyer can purchase at most one affiliated product within a submarket. This setup explicitly assumes that affiliated products in the same submarket are substitutable and that affiliated products from different submarkets are independent. In practice, a submarket can be defined on a case-by-case basis. Taking computer operating systems and software, for example, software products are typically grouped into educational software, editing software, simulation software, communication software, language software, multimedia software, etc. As for the e-reader and e-book system, e-books are typically grouped into mysteries, romances, science fiction, children's books, etc. In Section 4, I group sports video games into football games, baseball games, etc. To examine the substitutability of sports games, I perform several regressions and provide the results in Section A of the online appendix. The regression results indicate that games grouped in the same submarket are strongly substitutable, while games grouped in different submarkets are weakly substitutable. Nair (2007) and Lee (2013) assume no substitutability among video games, which is equivalent to defining each affiliated product as a submarket.

Below I first describe buyers' dynamic adoption of platforms and dynamic purchase of affiliated products. Next, I describe sellers' dynamic entry and pricing. Finally, I define an equilibrium of the model in which buyers and sellers behave optimally in each period given their information and beliefs, and these beliefs are in turn consistent with the actions of all agents.

2.1. Buyer Side

On the buyer side of the market, there is a discrete finite number of buyer types in the population, each indexed by i , $i = 1, \dots, I$. In each period of time, buyers who have not adopted any platform make their adoption decisions of platforms, and buyers who have already adopted a platform make their purchase decisions of affiliated products.

2.1.1. Platform Adoption. A platform's value depends on the availability, quality, and prices of its affiliated products. Let Γ_{ilt} denote a type i buyer's expected value of optimally purchasing products affiliated with platform l at time t . The functional form of Γ_{ilt} is derived from the affiliated product purchase portion of the model, which will be specified in Section 2.1.2. The expected lifetime utility that a type i buyer can obtain from adopting platform l at time t is

$$U_{ilt} = \alpha \Gamma_{ilt} + X_{lt} \gamma_i + \zeta_{lt} + \varepsilon_{ilt},$$

where α reflects how important affiliated products are in buyers' valuation of a platform; X_{lt} is a vector of observable platform characteristics including platform price, platform-specific fixed effects, time effects, and other time-varying platform characteristics; γ_i represents type i buyer's heterogeneous preference for observable platform characteristics; ζ_{lt} represents additional platform characteristics observed by buyers but not by researchers; and ε_{ilt} is an individual platform-time-specific utility shock.

Since platform intermediaries are durable goods, buyers are forward looking when they make the adoption decisions. The no-purchase option captures the value of delaying an adoption to a future period. I specify the utility of not adopting at time t as the sum of the discounted expected value of waiting and an individual time-specific utility shock:

$$U_{i0t} = \beta_1 E_t \left[\max \left\{ \max_l U_{il,t+1}, U_{i0,t+1} \right\} \right] + \varepsilon_{i0t},$$

where β_1 is buyer's discount factor in the platform market, the expectation is taken with respect to the distribution of future variables unknown to buyers conditional on the current information, and ε_{i0t} captures the utility of the outside good. As usual in the literature, ε_{ilt} and ε_{i0t} are assumed to follow the type I extreme value distribution demeaned by Euler's constant and are independently and identically distributed (i.i.d.) across platforms and buyer types and over time.

Let $\mathbf{S}_t$ denote the information set that affects buyers' platform adoption decision at time t . Then, a type i buyer's dynamic optimization problem can be written as

$$H_{it}(\varepsilon_{it}, \mathbf{S}_t) = \max \left\{ \max_l U_{ilt}, \varepsilon_{i0t} + \beta_1 E_t [E_\varepsilon H_{i,t+1}(\varepsilon_{i,t+1}, \mathbf{S}_{t+1}) | \varepsilon_{it}, \mathbf{S}_t] \right\},$$

where $H_{it}(\varepsilon_{it}, \mathbf{S}_t)$ is type i buyer's value function with information set $\mathbf{S}_t$ and utility shocks ε_{it} . Let $H_{it}(\mathbf{S}_t)$ denote the expected value function, that is, the value function before buyers know their utility shocks ε_{it} , that is, $H_{it}(\mathbf{S}_t) = \int_\varepsilon H_{it}(\varepsilon_{it}, \mathbf{S}_t) dF_\varepsilon(\varepsilon_{it})$. The integration with respect to the extreme value error terms has a closed form, and the deterministic component of buyer's value function satisfies

$$H_{it}(\mathbf{S}_t) = \ln \left\{ \sum_l \exp(\alpha \Gamma_{ilt} + X_{lt} \gamma_i + \zeta_{lt}) + \exp[\beta_1 E_t H_{i,t+1}(\mathbf{S}_{t+1} | \mathbf{S}_t)] \right\}. \quad (1)$$

Then, the probability of a type i buyer adopting platform l at time t is

$$B_{ilt}(\mathbf{S}_t) = \frac{\exp(\alpha \Gamma_{ilt} + X_{lt} \gamma_i + \zeta_{lt})}{\exp[\beta_1 E_t H_{i,t+1}(\mathbf{S}_{t+1} | \mathbf{S}_t)] + \sum_{l'} \exp(\alpha \Gamma_{il't} + X_{l't} \gamma_i + \zeta_{l't})}. \quad (2)$$

Let N_{it} denote the number of type i buyers who have not adopted any platform at time t . Then, the demand for platform l at time t is

$$Q_{lt} = \sum_i N_{it} B_{ilt}.$$

Notice that a buyer is assumed to adopt at most one platform in her life time, and once she adopts a platform, she is no longer active in the platform market. Hence, in this dynamic discrete choice demand model, $\{N_{it}\}_{t=1}^T$ evolves according to

$$N_{i,t+1} = N_{it} \left(1 - \sum_l B_{ilt} \right).$$

2.1.2. Affiliated Product Purchase. The affiliated product market consists of a finite number of separate submarkets, indexed by $m = 1, \dots, M$. Once a buyer adopts a platform, she becomes active in all submarkets of products affiliated with that platform. Below I describe buyers' demand in one submarket, and their demand in other submarkets are the same.

Utility. Let $\mathcal{J}_{mt}$ denote the set of affiliated products available for buyers to purchase in submarket m at time t . A type i buyer's lifetime expected utility from purchasing an affiliated product $j \in \mathcal{J}_{mt}$ at time t (provided she has already adopted the compatible platform) is

$$u_{ijt} = x_{jt} \psi + \varphi_i p_{jt} + \xi_{jt} + \epsilon_{ijt},$$

where x_{jt} is a vector of covariates including platform-specific fixed effects, time effects, and other observed product characteristics; p_{jt} is the product price; ξ_{jt} is additional product characteristics observed by buyers

but not by researchers; and ϵ_{ijt} is individual product-time-specific utility shock. Here, φ_i measures type i buyers' marginal disutility from prices of affiliated products.

In the dynamic environment, the utility of not buying in the submarket m at time t is the sum of the discounted expected value of waiting and a utility shock:

$$u_{imt}^0 = \beta_2 E_t \left[\max \left\{ \max_{j \in \mathcal{J}_{m,t+1}} u_{ij,t+1}, u_{im,t+1}^0 \right\} \right] + \epsilon_{imt}^0,$$

where β_2 is the buyer's discount factor in the affiliated product market, and ϵ_{im0} is the type i buyer's idiosyncratic taste from not buying any product in submarket m . As usual, ϵ_{ijt} and ϵ_{imt}^0 are assumed to follow the type I extreme value distribution demeaned by Euler's constant and are i.i.d. across products and buyer types and over time.

Purchase Decision. Let $\mathbf{s}_{mt}$ denote the information set affecting agents' choices in submarket m at time t , and let $G_{it}(\mathbf{s}_{mt}, \mathbf{p}_{mt})$ denote the type i buyer's value function before they know their demand shocks ϵ_{mit} . Because of the specific functional form of the utility shocks, the value function can be written as

$$G_{it}(\mathbf{s}_{mt}, \mathbf{p}_{mt}) = \ln \left\{ \sum_{j \in \mathcal{J}_{mt}} \exp(x_{jt}\psi + \varphi_i p_{jt} + \xi_{jt}) + \exp[\beta_2 E_t G_{i,t+1}(\mathbf{s}_{m,t+1}, \mathbf{p}_{m,t+1} | \mathbf{s}_{mt}, \mathbf{p}_{mt})] \right\}. \quad (3)$$

The probability that a type i buyer purchases affiliated product $j \in \mathcal{J}_{mt}$ at time t is

$$b_{ijt}(\mathbf{s}_{mt}, \mathbf{p}_{mt}) = \exp(x_{jt}\psi + \varphi_i p_{jt} + \xi_{jt}) / \left\{ \sum_{j' \in \mathcal{J}_{mt}} \exp(x_{j't}\psi + \varphi_i p_{j't} + \xi_{j't}) + \exp[\beta_2 E_t G_{i,t+1}(\mathbf{s}_{m,t+1}, \mathbf{p}_{m,t+1} | \mathbf{s}_{mt}, \mathbf{p}_{mt})] \right\}. \quad (4)$$

The demand for product $j \in \mathcal{J}_{mt}$ at time t is

$$q_{jt}(\mathbf{s}_{mt}, \mathbf{p}_{mt}) = \sum_i n_{mit} b_{ijt}(\mathbf{s}_{mt}, \mathbf{p}_{mt}),$$

where n_{mit} is the number of type i buyers who are active in submarket m at time t . Notice that each buyer is assumed to buy at most one affiliated product in a submarket. Under this assumption, a buyer is no longer active in a submarket once she has made a purchase in that submarket. Meanwhile, new buyers enter a submarket once they adopt the compatible platform. Therefore, the evolution of $\{n_{mit}\}_{t=1}^T$ follows

$$n_{mi,t+1} = n_{mit} \left(1 - \sum_{j \in \mathcal{J}_{mt}} b_{ijt} \right) + n_{mit}^e, \quad (5)$$

where $n_{mit}(1 - \sum_{j \in \mathcal{J}_{mt}} b_{ijt})$ is the mass of buyers who do not buy in period t and remain active in the next

period, and $n_{mit}^e = N_{it} B_{ilt}$ is the mass of new buyers who adopt the compatible platform l , as described in Section 2.1.1.³ Notice that the mass of buyers remaining in a submarket is endogenous to the historic entry and pricing behavior of all sellers in that submarket. The dynamics of entry and pricing introduce a dynamic evolution of the buyer distribution in submarket m .

Total Utility from Affiliated Products. In Section 2.1.1, I have specified that buyers' valuation of a platform depends on the total utility from being able to purchase the affiliated products, Γ_{ilt} . Notice that a buyer who adopts a platform starts to buy the affiliated products in the next period. Given that the affiliated product submarkets are assumed to be independent, Γ_{ilt} is the sum of those expected values of being active in all submarkets affiliated with platform l starting from time $t + 1$ until the end time period T ,

$$\Gamma_{ilt} = E_t \left[\sum_{\tau=t+1}^T \beta_2^\tau \sum_{m \in M_l} G_{i\tau}(\mathbf{s}_{m\tau}, \mathbf{p}_{m\tau}) \right], \quad (6)$$

where M_l is the set of submarkets affiliated with platform l .

2.2. Seller Side

On the seller side, in each time period and in each submarket, incumbent sellers optimally choose their prices, and potential entrants optimally decide whether or not to release a new product.

2.2.1. An Incumbent Seller's Problem. Let c_l denote the unit cost of affiliated products of platform l , including the production cost and the royalty fee paid to platform l . Both of the two costs are time invariant and platform specific, which has been documented by previous studies on various two-sided markets.⁴

An incumbent seller's one-period profit depends on its own price choice this period (p_{jt}) and its competitors' prices ($\mathbf{p}_{-jt}$); moreover, it also depends on the state vector $\mathbf{s}_{mt}$ in submarket m including the set of available products, product characteristics, and the distribution of active buyers. An incumbent's optimization problem is to choose a price to maximize its discounted profit,

$$\begin{aligned} \Pi_{jt}(\mathbf{s}_{mt}, p_{jt}, \mathbf{p}_{-jt}) &= E_t [\pi_{jt}(\mathbf{s}_{mt}, p_{jt}, \mathbf{p}_{-jt})] \\ &+ E_t \left\{ \sum_{\tau=t+1}^T \beta_3^{\tau-t} \left[\max_{p_{j\tau}} \pi_{j\tau}(\mathbf{s}_{m\tau}, p_{j\tau}, \mathbf{p}_{-j\tau}) \right] \middle| \mathbf{s}_{mt}, \mathbf{p}_{mt} \right\}, \end{aligned}$$

where $\pi_{jt}(\mathbf{s}_{mt}, p_{jt}, \mathbf{p}_{-jt}) = (p_{jt} - c_l) q_j(\mathbf{s}_{mt}, \mathbf{p}_{mt})$ is seller j 's one-period profit, β_3 is incumbent sellers' discount factor, the first expectation is taken with respect to competing sellers' price choices in the current time period, and the second expectation is taken with respect to the distribution of future state variables and competing sellers' price choices in future periods.

2.2.2. A Potential Entrant's Problem. Every period, a set of $\mathcal{E}_{mt}$ potential entrants simultaneously choose whether or not to enter a submarket m . The entry cost of a potential entrant j is assumed to be $\lambda_l + v_{jt}$, where λ_l is the component that is common to all products affiliated with platform l , and v_{jt} is a private information shock that is assumed to be independently and identically distributed across sellers and periods with cumulative distribution function $F_v(\cdot)$. Each potential entrant $j \in \mathcal{E}_{mt}$ first draws an entry cost from a known distribution and then decides whether to enter. Potential entrants are short lived and base their entry decisions on the net present value of entering today; they do not take the option value of delaying entry into account. If a potential entrant enters, it pays the entry cost and starts to earn profit next period; if not, it earns zero profits. Let $y_{j,t+1} = 1$ denote that entrant j decides to enter at time t , and let $y_{j,t+1} = 0$ denote that entrant j decides not to enter. A potential entrant j 's optimization problem is to compare the entry cost and the expected profit. The optimal strategy is to enter if the expected profit exceeds the entry cost and not to enter otherwise.

2.2.3. Perceived Strategy Function. Because a potential entrant's entry decision depends on its own entry cost shock, which is unknown to buyers and other sellers, other agents cannot know exactly a potential entrant's entry strategy even if they can observe the actual outcomes. We can define a set of conditional choice probabilities for $j \in \mathcal{E}_{mt}$ such that

$$\rho_{jt}(\mathbf{s}_{mt}) = \int 1\{y_{j,t+1}(\mathbf{s}_{mt}, v_{jt}) = 1\} dF_v(v_{jt}),$$

where $1\{\cdot\}$ is an indicator function. The probabilities represent the expected behavior of entrant j from the point of view of buyers and the rest of the sellers. The game has a Markov structure, and I assume that each seller plays Markov strategies. In particular, if $\mathbf{s}_{mt} = \mathbf{s}_{m't}$, then seller j 's decisions in submarket m and m' are the same. Let $\Psi = \{\Psi_{jt}(\mathbf{s}_{mt})\}$ be a set of strategy functions or decision rules, one for each seller, with $\Psi_{jt}(\mathbf{s}_{mt}) = p_{jt}(\mathbf{s}_{mt})$ if j is an incumbent seller and $\Psi_{jt}(\mathbf{s}_{mt}) = \rho_{jt}(\mathbf{s}_{mt})$ if j is a potential entrant.

2.2.4. Incumbent's Bellman Equation. Let $V_{jt}(\mathbf{s}_{mt} | \Psi)$ denote the expected net present value of all future cash flows to incumbent seller $j \in \mathcal{J}_{mt}$ at state vector $\mathbf{s}_{mt}$, computed under the presumption that buyers respond optimally and other sellers follow their strategies in Ψ . By Bellman's principle of optimality, it can be written as

$$V_{jt}(\mathbf{s}_{mt} | \Psi) = \max_{\tilde{p}_{jt}} \pi_{jt}(\mathbf{s}_{mt}, \tilde{p}_{jt}, p_{-jt}) + \beta_3 E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, \tilde{p}_{jt}, \Psi_{-jt}], \quad (7)$$

where $E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, p_{jt}, \Psi_{-j}]$

$$= \int_{\mathcal{E}_{m,t+1}} \left[\sum_{y_{m,t+1}} V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) \times f_j(y_{m,t+1} | \mathbf{s}_{mt}, p_{jt}, \Psi_{-jt}) \right] d\xi_{m,t+1}$$

is the expected value function conditional on seller j choosing p_{jt} and the other sellers behaving according to Ψ . Here, the conditional transition probability function is

$$f_j(y_{m,t+1} | \mathbf{s}_{mt}, p_{jt}, \Psi_{-j}) = \prod_{k \in \mathcal{E}_{mt}} \rho_{kt}(\mathbf{s}_{mt})^{y_{k,t+1}} (1 - \rho_{kt}(\mathbf{s}_{mt}))^{1-y_{k,t+1}}. \quad (8)$$

The optimal pricing strategy in response to profile Ψ is the solution of the right-hand side of Equation (7), denoted as $p_{jt}(\mathbf{s}_{mt} | \Psi)$.

2.2.5. Entrant's Bellman Equation. Let $V_{jt}^e(\mathbf{s}_{mt}, v_{jt} | \Psi)$ denote the expected net present value of all future cash flows to potential entrant $j \in \mathcal{E}_{mt}$ at state vector $\mathbf{s}_{mt}$ and entry cost shock v_{jt} , computed under the presumption that buyers respond optimally and other sellers behave according to strategy profile Ψ :

$$V_{jt}^e(\mathbf{s}_{mt}, v_{jt} | \Psi) = \max_{\tilde{y}_{j,t+1} \in \{0,1\}} \tilde{y}_{j,t+1} \{-\lambda_l - v_{jt} + \beta_4 E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, \Psi]\},$$

where β_4 is entrants' discount factor and

$$E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, \Psi] = \int_{\mathcal{E}_{m,t+1}} \left[\sum_{y_{m,t+1}} V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) f_j(y_{m,t+1} | \mathbf{s}_{mt}, \Psi) \right] d\xi_{m,t+1}$$

is the expected value function conditional on seller j choosing entering and the other sellers behaving according to strategy profile Ψ . Here, the conditional transition probability function is given by

$$f_j(y_{m,t+1} | \mathbf{s}_{mt}, \Psi) = \prod_{k \in \mathcal{E}_{mt}, k \neq j} \rho_{kt}(\mathbf{s}_{mt})^{y_{k,t+1}} (1 - \rho_{kt}(\mathbf{s}_{mt}))^{1-y_{k,t+1}}, \quad (9)$$

where the j th dimension of $\mathbf{y}_{m,t+1}$ equals one.

The optimal entry decision follows a cutoff rule characterized by

$$y_{j,t+1}(\mathbf{s}_{mt}, v_{jt} | \Psi) = \begin{cases} 1 & \text{if } v_{jt} \leq \bar{v}_{jt}(\mathbf{s}_{mt} | \Psi), \\ 0 & \text{otherwise,} \end{cases}$$

where

$$\bar{v}_{jt}(\mathbf{s}_{mt} | \Psi) = \beta_4 E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, \Psi] - \lambda_l$$

is the cutoff entry cost shock for which the potential entrant is indifferent between entering and staying out of the submarket. Then, the probability of entering is

$$\rho_{jt}(\mathbf{s}_{mt} | \Psi) = \int I[v_{jt} \leq \bar{v}_{jt}(\mathbf{s}_{mt} | \Psi)] dF_v(v_{jt}) = F_v[\bar{v}_{jt}(\mathbf{s}_{mt} | \Psi)].$$

Thus, the unconditional Bellman equation of a potential entrant j can be written as

$$V_{jt}^e(\mathbf{s}_{mt} | \Psi) = \max_{\bar{\rho}_{jt}} - \int_{v_{jt} < F_v^{-1}(\bar{\rho}_{jt})} v_{jt} dF_v(v_{jt}) + \bar{\rho}_{jt} \{-\lambda_l + \beta E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \Psi) | \mathbf{s}_{mt}, \Psi]\}. \quad (10)$$

2.3. Equilibrium

I consider the pure-strategy Markov perfect equilibrium (MPE) concept. The MPE of this model is defined by a set of value functions, $\{G_{it}(\mathbf{s}_{mt}, \mathbf{p}_{mt}^*)\}$, $H_{it}(\mathbf{S}_t)\}_{i=1}^I$ and $\{V_{jt}(\mathbf{s}_{mt})\}_{j \in \mathcal{J}_{mt}}$, a set of price functions, $\{p_{jt}^*(\mathbf{s}_{mt})\}_{j \in \mathcal{J}_{mt}}$, and a set of entry functions, $\{\rho_{jt}^*(\mathbf{s}_{mt})\}_{j \in \mathcal{E}_{mt}}$, such that Equations (1)–(10) are simultaneously satisfied at every state $\mathbf{s}_{mt}$. In other words, the equilibrium is a fixed point of the model defined by Equations (1)–(10), with the following properties.

(i) *Sellers.* Equation (7) implies that at equilibrium, when faced with state $\mathbf{s}_{mt}$, each incumbent seller's pricing policy is a best response to other sellers' strategies and buyers' reaction at that state. Meanwhile, Equation (10) implies that at equilibrium, when faced with state $\mathbf{s}_{mt}$, each entrant's entry policy is a best response to other sellers' strategies and buyers' reaction at that state.

(ii) *Buyers.* Equations (3) and (4) imply that when faced with a state $\mathbf{s}_{mt}$ and price $p(\mathbf{s}_{mt})$, buyers who have adopted a platform rationally anticipate sellers' future pricing and entry in the affiliated product market, and optimally make purchase decisions of affiliated products. At the same time, Equations (1) and (2) imply that buyers who have not adopted any platform yet make adoption decisions of platforms by maximizing intertemporal utility. In addition, the consumption value of a platform is given by the Equation (6).

(iii) *State transition.* Sellers take into account the effect of their actions on the evolution of states. The transition of the buyer distribution follows Equation (5). In the eyes of incumbent sellers, the transition of the product availability follows Equation (8), and in the eyes of potential entrants who decide to enter, it follows Equation (9).

The model has a finite horizon and hence I solve it using backward deduction. At the last period ($t = T$), the subgame is a static game that is basically a static Bertrand price competition game because no entrants would decide to enter at $t = T$. I numerically compute the equilibrium prices for this static game. Folding back, the subgame at time $T - 1$ becomes a one-period game, and I numerically compute the fixed point of the best response system given the equilibrium prices

at $t = T$. Then, I proceed to time $T - 2$ and so on. The next section describes the details of the algorithm of computing the equilibrium and estimating the model. The algorithm is programmed in Fortran 95. I repeat the algorithm not only for different initial values of the value functions and perceived equilibrium strategies, but also for different starting values of parameters. The algorithm converges smoothly to almost the same results.

Notice that the main focus of this paper is the dynamic strategic interaction between the two sides of the market within one generation of platforms, and hence the model has a finite horizon. However, if we study multiple generations, buyers may replace their old-generation platforms with a new-generation platform, and hence an infinite-horizon model might be more appropriate. In this case, the distribution of active buyers does not change since buyers do not exit. A buyer's trade-off is typically between waiting for a lower price or higher quality product, and either forgoing consumption until then or purchasing a product now and retiring it earlier than needed. We can follow Gowrisankaran and Rysman (2012) to model the buyer side. Since a buyer can hold several varieties of the products and the utilities from the different varieties interact with each other, the state variable includes a vector describing the buyer's current holdings, and, moreover, buyers who previously purchased a product have a different value of no purchase since their alternative is to stay with their current products. To reduce the dimension of the holdings variable, we can assume that a buyer only holds a single option at any point in time so that the holding vector becomes a scalar. Then, we can still apply the estimation procedure in Section 3.

3. Bayesian Estimation

Let θ denote the vector of parameters in the model that need to be estimated. Let *data* denote all the data available for estimation, which include two parts: (i) the prices and quantity sold of each platform in each time period and (ii) the availability, characteristics, prices, and quantity sold of each affiliated product in each time period across M submarkets. Hence, $\text{data} = \{\mathbf{P}_t, \mathbf{Q}_t, \{\mathbf{y}_{mt}, \mathbf{x}_{mt}, \mathbf{p}_{mt}, \mathbf{q}_{mt}\}_{m=1}^M\}_{t=1}^{T_d}$, where T_d is the number of time periods in the data set. I assume that the data are generated from the model presented in Section 2.

3.1. Posterior

Let $\mathcal{L}(\text{data} | \theta)$ denote the likelihood. Rather than using the maximum likelihood estimation method, I employ the Bayesian MCMC method to sample the parameter vector θ from its posterior distribution,

$$\mathcal{P}(\theta | \text{data}) \propto \mathcal{L}(\text{data} | \theta)\pi(\theta), \quad (11)$$

where $\pi(\theta)$ is the prior distribution of the parameter vector θ .

3.2. Likelihood Contributions

The platform adoption is a dynamic discrete choice model. Since the platform utility equation includes platform fixed effects and time effects, ζ_{lt} captures only those temporary components of a platform's unobserved quality. To specify the likelihood contribution of the demand for platforms, I follow Jiang et al. (2009) and assume that ζ_{lt} are normally distributed with mean zero and variance σ_ζ^2 , independent across products and over time. The distribution of the aggregate demand shocks generates the distribution of platform sales in each time period. Conditional on the state, the joint density of the sales of all platforms at time t is

$$\mathcal{L}_Q(\mathbf{Q}_t | \cdot; \theta) = \prod_l [\phi(\zeta_{lt}/\sigma_\zeta)/\sigma_\zeta] |J_{(\mathbf{Q}_t \rightarrow \zeta_{lt})}^{-1}|, \quad (12)$$

where $\phi(\cdot)$ is the probability density function of the standard normal distribution, and $J_{(\mathbf{Q}_t \rightarrow \zeta_{lt})}$ is the Jacobian matrix.

Assume that the log of an affiliated product j 's observed price $\ln(p_{jt})$ is the sum of the log of pricing policy function $\ln(p_j^*(\mathbf{s}_{mt}; \theta))$ and a stochastic shock ς_{jt} :

$$\ln(p_{jt}) = \ln(p_j^*(\mathbf{s}_{mt}; \theta)) + \varsigma_{jt}.$$

The demand and supply shocks are assumed to be normally distributed independent across time periods and markets:

$$\begin{bmatrix} \xi_{mt} \\ \varsigma_{mt} \end{bmatrix} = \begin{bmatrix} \Omega^{-1}(\mathbf{s}_{mt}, \mathbf{q}_{mt}; \theta) \\ \ln(\mathbf{p}_{mt}) - \ln(p^*(\mathbf{s}_{mt}; \theta)) \end{bmatrix} \\ \sim N\left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \Omega \equiv \begin{bmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{bmatrix}\right).$$

The multivariate normal distribution of shocks induces the joint density of the sales and price:

$$\begin{aligned} \mathcal{L}_{q,p}(\mathbf{q}_{mt}, \mathbf{p}_{mt} | \cdot; \theta) \\ = \frac{1}{(2\pi)^J |\Omega|^{1/2}} \exp\left(-\frac{1}{2} [\xi_{mt} \varsigma_{mt}] \Omega^{-1} [\xi_{mt} \varsigma_{mt}]'\right) \\ \cdot |J_{(\mathbf{q}_{mt}, \mathbf{p}_{mt}) \rightarrow (\xi_{mt}, \varsigma_{mt})}^{-1}|. \end{aligned} \quad (13)$$

The derivation of the Jacobian $J_{(\mathbf{q}_{mt}, \mathbf{p}_{mt}) \rightarrow (\xi_{mt}, \varsigma_{mt})}$ is provided in Section B of the online appendix.

To specify the likelihood contribution of potential entrants' entry actions, I assume that the entry cost shocks follow an independent normal distribution with mean zero and variance σ_v^2 .⁵ Hence, conditional on the state vector $\mathbf{s}_{mt}$, the likelihood contribution of entrant $j \in \mathcal{E}_{mt}$ is

$$\begin{aligned} \mathcal{L}_y(y_{j,t+1} | \cdot; \theta) \\ = \left(\Phi \left[\frac{\beta_4 E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \mathbf{s}_{mt}; \theta)] - \lambda_l}{\sigma_v} \right] \right)^{y_{j,t+1}} \\ \times \left(1 - \Phi \left[\frac{\beta_4 E_t[V_{j,t+1}(\mathbf{s}_{m,t+1} | \mathbf{s}_{mt}; \theta)] - \lambda_l}{\sigma_v} \right] \right)^{1-y_{j,t+1}}. \end{aligned} \quad (14)$$

The likelihood, therefore, is

$$\mathcal{L}(\text{data} | \theta) = \prod_{t=1}^{T_d} \left\{ \mathcal{L}_Q(\mathbf{Q}_t | \cdot; \theta) \prod_{m=1}^M \left[\mathcal{L}_{q,p}(\mathbf{q}_{mt}, \mathbf{p}_{mt} | \cdot; \theta) \cdot \prod_{j \in \mathcal{E}_{mt}} \mathcal{L}_y(y_{j,t+1} | \cdot; \theta) \right] \right\}.$$

3.3. Estimation Algorithm and Computation

The estimation procedure involves two loops: in the outer loop, I use the Metropolis–Hastings algorithm to draw a sequence of parameter vectors from their posterior distributions, and in the inner loop, for a given parameter vector along the MCMC chain, I solve the dynamic model using backward deduction, combining the kernel approximation and interpolation approaches.

3.3.1. Outer Loop: Metropolis–Hastings Step. The posterior distribution in Equation (11) is a high-dimensional and complex function of the parameters. It is known that instead of drawing the entire parameter vector at once, it is often simpler to partition it into blocks and draw the parameters of each block separately given the other parameters.⁶ Parameters of the model are partitioned into three blocks: (i) the first block includes all parameters directly affecting buyers' adoption decisions of platforms, i.e., the parameters in the platform utility function, $\theta_1 = (\alpha, \gamma, \sigma_\zeta)$; (ii) the second block includes all parameters directly affecting buyers' purchase decisions of affiliated products and incumbent sellers' pricing decisions, i.e., the parameters in the utility function of affiliated products, $\theta_2 = (\psi, \varphi, c, \Omega)$; and (iii) the last block includes all parameters directly affecting entrants' entry decisions, i.e., the mean and the standard deviation of sellers' entry cost to each platform, $\theta_3 = (\lambda, \sigma_v)$.

Consider a particular MCMC iteration k . For each block r , a candidate parameter vector $\tilde{\theta}_r^{(k)}$ is drawn from a proposed density $f_r(\tilde{\theta}_r^{(k)} | \theta_r^{(k-1)})$. A random-walk Metropolis chain is used for the draw of $\tilde{\theta}_r^{(k)}$, that is,

$$\tilde{\theta}_r^{(k)} = \theta_r^{(k-1)} + \text{MVN}(0, \kappa \Sigma_r),$$

where Σ_r is the candidate covariance matrix, and κ is a scaling constant. Then, the candidate parameter vector $\tilde{\theta}_r^{(k)}$ is accepted with probability

$$\tilde{\eta}_r^{(k)} = \min \left\{ 1, \frac{\mathcal{L}_r(\cdot | \tilde{\theta}_r^{(k)}, \theta_r^{(k-1)}) f_r(\tilde{\theta}_r^{(k)} | \theta_r^{(k-1)}) \pi_r(\tilde{\theta}_r^{(k)})}{\mathcal{L}_r(\cdot | \theta_r^{(k-1)}, \theta_r^{(k-1)}) f_r(\theta_r^{(k-1)} | \tilde{\theta}_r^{(k)}) \pi_r(\theta_r^{(k-1)})} \right\}. \quad (15)$$

3.3.2. Inner Loop: Fixed Point Step. Evaluating the likelihood in (15) requires solving for the equilibrium of the dynamic model for any given parameter vector θ . The computation difficulty comes in two parts. One part is computing each agent's value function for a given a strategy profile, that is, solving for the

fixed points of a single-agent dynamic programming problem. The other part is computing the equilibrium strategy profile, that is, solving for the fixed points of the best response system. To reduce the computational burden, I develop an estimation procedure that solves the fixed point of a dynamic model in conjunction with the Bayesian MCMC estimation.

Nonparametric Approximation of Value Function. Conventional estimation methods iterate the Bellman operator until convergence, which is computationally difficult for relatively complicated models. I follow IJC's approach, which approximates the expected value function using the weighted average of value functions of most recent iterations. Since the model is a finite-period model that is nonstationary, I compute and store the value functions for each period and approximate the expected value functions in period t using the set of value functions in period $t + 1$ (also see Ishihara and Ching 2012). Incumbent seller j 's value function in iteration k is approximated as

$$\hat{V}_{jt}^{(k)}(\mathbf{s}_{mt}, \theta) = \sum_{n=1}^{N(k)} V_{jt}^{(k-n)}(\mathbf{s}_{mt}, \tilde{\theta}^{(k-n)}) \times \frac{K_h(\theta - \tilde{\theta}^{(k-n)})}{\sum_{n=1}^{N(k)} K_h(\theta - \tilde{\theta}^{(k-n)})}, \quad (16)$$

where $K_h(\cdot)$ is a multivariate kernel with bandwidth $h > 0$, and $V_{jt}^{(k)}(\mathbf{s}_{mt}, \theta)$ is seller j 's value function at state $\mathbf{s}_{mt}$ conditional on all other sellers playing the equilibrium $\hat{\Psi}^{(k)}$:

$$V_{jt}^{(k)}(\mathbf{s}_{mt}, \theta) = \max_{\tilde{p}_{jt}} (\tilde{p}_{jt} - c_j) q(\tilde{p}_{jt}, \hat{p}_{-jt}^{(k)}) + \beta_3 E_t [\hat{V}_{j,t+1}^{(k)}(\mathbf{s}_{mt+1}, \theta) | \mathbf{s}_{mt}]. \quad (17)$$

Store the solved value functions $V_{jt}^{(k)}$ for future MCMC iterations. The approximated value function given by Equation (16) is the weighted average of the value functions of the $N(k)$ most recent iterations. IJC show that as the MCMC iterations and the number of past iterations for approximating the value functions increase, the value function converges to the true value functions, and the posterior parameter draws based on the value functions converge to the true posterior distributions.

Nonparametric Approximation of Policy Function. A widely used approach is the nested fixed point (NFP) approach that computes the equilibrium by iterating the best response operator until convergence. However, it is too computationally demanding for the model described in Section 2. As an alternative, I propose to use a kernel method to approximate the policy function at current iteration using the policy functions of the past iterations in which the parameter vector is "close" to the current parameter vector. Seller j 's policy

function at time t that is perceived by other agents in iteration k is computed as

$$\hat{\Psi}_{jt}^{(k)}(\mathbf{s}_{mt}, \theta) = \sum_{n=1}^{N(k)} \Psi_{jt}^{(k-n)}(\mathbf{s}_{mt}, \tilde{\theta}^{(k-n)}) \times \frac{K_h(\theta - \tilde{\theta}^{(k-n)})}{\sum_{n=1}^{N(k)} K_h(\theta - \tilde{\theta}^{(k-n)})}, \quad (18)$$

where $\Psi_{jt}^{(k)}$ is the policy function at time t in the iteration k . For incumbent seller j , the optimal price is the solution to the incumbent's optimization problem in Equation (7) given other agents playing $\hat{\Psi}_{jt}^{(k)}$, denoted as $p_{jt}^{(k)}(\mathbf{s}_{mt}, \theta)$. For entrant j , the equilibrium entry probability is the solution to the potential entrant's optimization problem in Equation (10) given other agents playing $\hat{\Psi}_{jt}^{(k)}$. Under the assumption of normally distributed entry cost shocks, it is

$$\rho_{jt}^{(k)}(\mathbf{s}_{mt}, \theta) = \Phi(\{\beta_4 E_t [\hat{V}_{j,t+1}^{(k)}(\mathbf{s}_{mt+1}, \theta) | \mathbf{s}_{mt}] - \lambda_l\} / \sigma_v).$$

In essence, the perceived policy functions are approximated by the weighted average of policy functions of past iterations. Store the solved policy functions, $p_{jt}^{(k)}$ and $\rho_{jt}^{(k)}$, for future MCMC iterations. Basically, I combine the Metropolis–Hastings step with the fixed point step for a single iteration. I do not fully solve for the equilibrium of the dynamic model but use the results from previous MCMC iterations. This spirit is similar to Aguirregabiria and Mira's (2007) nested pseudolikelihood method that gradually updates the equilibrium probabilities and recursively obtains the estimators.

Interpolation. However, to obtain the expected value functions in Equation (17), we still need to compute Equations (16) for every possible point of the state space. Because of the "curse of dimensionality," it is computationally burdensome to achieve it even with the nonparametric approximation method described above. To cope with this problem, I adopt an interpolation method of Keane and Wolpin (1994) (also see Crawford and Shum 2005). In the first step, I randomly choose a subset of the state points every period and compute the values at those points with the nonparametric approximation approach described above. Next, I interpolate the value functions with a fourth-order polynomial in that subset. Last, for each current state, I simulate a next-period state using the approximated policy functions and then use the predicted value at that simulated next-period state as the continuation value. In practice, I simulate the next-period state for a finite number of times and then take the average of the predicted values. This procedure significantly alleviates the computational burden and makes it possible to estimate models with large state spaces and rich structure.

3.4. Monte Carlo Exercises

To assess the relative performance of the proposed estimation method versus conventional methods, I estimate the dynamic demand model of Dube et al. (2012) using four different estimation algorithms. Details about the model and estimation algorithm are provided in Section C of the online appendix.

For the first two algorithms, during each MCMC step, I use the kernel approximation to obtain the expected value functions and perceived policy functions. To cope with the state space problem, I use the interpolation method with a fourth-order polynomial and four interpolation nodes in the first algorithm, and I use Rust's (1997) random grid approximation with 100 grid points in the second algorithm. For the third and fourth algorithms, during each MCMC step, I iterate the Bellman operator and BLP contraction mapping (Berry et al. 1995) until convergence. I use the interpolation approach and random grid approximation approach for the third and fourth algorithms, respectively.

I conducted 20 replications in this experiment and used the last 10,000 draws from the posterior distribution to derive the posterior means and standard deviations. Table 1 reports the average of the posterior means ($\overline{PM}$) and the average of the posterior standard deviations ($\overline{PSD}$) for the 20 replications. The kernel approximation algorithm is considerably faster than the NFP algorithm, and the interpolation method dramatically reduces the estimation time. A typical run of the kernel approximation algorithm requires 0.56 hours with the polynomial interpolation approach and 3.04 hours with the random grid approach, whereas the

NFP algorithm requires 8.06 hours with the interpolation approach and 25.17 hours with the random grid approach.

3.5. Identification

This subsection provides an informal discussion of identification in this model. Platform characteristic coefficients and affiliated product characteristic coefficients are identified from variation in sales as such characteristics change. Price coefficients (γ_i and φ_i) and the initial share of each buyer type are identified from variation in sales as the price changes within and across submarkets and the rate of change in sales over time in response to price change.⁷

The parameter α captures the impact of affiliated products' availability, prices, and characteristics on a buyer's valuation of a platform. Affiliated products' availability and prices are dependent on platform adoption, and hence Γ_{it} is endogenous. The identification of α relies on the exogenous assumption of affiliated product characteristics. It is a concern that affiliated product attributes x_{jt} might be correlated with the unobserved attributes of platform ζ_{it} . For example, affiliated product sellers might invest more in creating better products for a platform that experiences positive demand shocks, and as a result, x_{jt} would be correlated with ζ_{it} . Since the platform utility equation includes the platform dummy and time dummy variables, ζ_{it} captures only those transitory components of the unobserved quality of a platform. The implicit assumption is that those temporary shocks ζ_{it} cannot be expected by the sellers in advance.

The platform-specific unit costs of affiliated products (c_i) are identified from the average prices of

Table 1. Monte Carlo Results for a Dynamic BLP Model

Parameters	True	Kernel approx. (interpolation)		Kernel approx. (random grid)		NFP (interpolation)		NFP (random grid)	
		$\overline{PM}$	$\overline{PSD}$	$\overline{PM}$	$\overline{PSD}$	$\overline{PM}$	$\overline{PSD}$	$\overline{PM}$	$\overline{PSD}$
$\alpha_{j=1}^x$	4	4.043	0.339	4.031	0.389	4.062	0.320	4.062	0.285
$\alpha_{j=2}^x$	3	3.075	0.349	3.025	0.350	3.050	0.308	3.022	0.345
$\alpha_{i=1}^b$	-1	-1.007	0.023	-0.991	0.023	-1.003	0.024	-1.006	0.022
$\alpha_{i=2}^b$	-2	-2.147	0.335	-2.094	0.442	-2.136	0.179	-2.006	0.228
w_1	0.7	0.693	0.060	0.701	0.085	0.698	0.060	0.700	0.057
η_0	0.2	0.206	0.057	0.200	0.064	0.209	0.046	0.206	0.040
η_1	0.8	0.787	0.020	0.791	0.020	0.796	0.020	0.793	0.019
η_2	0	0.003	0.019	0.003	0.019	0.004	0.018	0.001	0.018
$e^{\gamma\xi}$	1	0.996	0.033	1.001	0.033	1.006	0.030	1.003	0.031
$\gamma_{\omega\xi}$	0.5	0.505	0.048	0.496	0.053	0.503	0.050	0.502	0.046
$e^{\gamma\omega}$	0.866	0.860	0.029	0.866	0.029	0.861	0.026	0.867	0.029
MCMC iterations		50,000		30,000		30,000		20,000	
CPU time (hours)		0.56		3.04		8.06		25.17	

Notes. For kernel approximation, I use the past 1,000 iterations to approximate the expected value functions and common demand shocks at the current iteration. For the NFP algorithm, I use the tolerance 10^{-14} for both the Bellman iteration and the BLP contraction mapping iteration, and I use Gauss-Hermite quadrature with three nodes to compute the numerical integration of future states. The generated data have $T = 50$ time periods and $M = 5$ distinct markets. Each market has two competing products, $J = 2$. There are two types of consumers, $I = 2$, with common discount factor $\beta = 0.99$.

affiliated products when they exit the market. The argument is that affiliated products are being sold only if their prices are higher than the unit cost. The platform-specific entry costs for affiliated product sellers (λ_i) are identified from the entry frequencies conditional on the expected profits from entering that are computed from the product competition market.

4. Application: The U.S. Video Game Industry

In this section, I apply the model specified in Section 2 and the estimation method presented in Section 3 to the U.S. video game industry. The video game industry is a two-sided market in which consoles (platforms) act as platform intermediaries, and buyers and sellers of video games (affiliated products) are on the two sides of the market. On the buyer side, console providers design and sell consoles to buyers, who pay a one-time fixed fee for the console that allows them to purchase and consume the affiliated games. On the seller side, console providers charge third-party game sellers royalties for the rights to the code that allows game sellers to make their games compatible with the console. The royalty fee is not a one-time payment; rather, it is a unit payment for each copy that they sell to buyers. In fact, console makers manufacture all the video games themselves so they can track sales for royalty collection. Console makers also develop and publish video games for their own platforms; in-house game titles do not need to pay royalty fees to console makers. I treat their prices and availabilities as given to other third-party game sellers.

To satisfy buyers' needs for the latest technology, console providers have introduced new systems approximately every five years. The data used in this paper cover the 32/64-bit generation, or the fifth generation, of the U.S. video game market. The data include three specific consoles: Sega Saturn, released in May 1995; Sony PlayStation, released in September 1995; and Nintendo 64, released in September 1996. Among them, Sega, a very successful incumbent for many years in this industry, failed to launch its platform and exited the industry.

4.1. Data

The main data set was obtained from the NPD Group, a market research firm. The data include the monthly revenue and unit sales of the three consoles, Sega Saturn (Saturn), Sony PlayStation (PS), and Nintendo 64 (N64), from May 1995 through February 2002. Sony was a new entrant to this industry, and the PS soon became the leading platform, taking around 60% of the market. Nintendo was the major competitor of Sony and had a market share of 37%. Sega ran a distant third behind the other two and actually stopped producing in 1998. I take the ratio of revenue over unit

sales in each month to calculate the console price. Since the sixth generation started when Sony launched its PlayStation 2 (PS2) in October 2000, the data set covers the entire fifth-generation video game industry. Additionally, none of the consoles was backwardly compatible, eliminating the concern that a previously existing buyer base might have given one console platform an advantage.

The data set also includes the monthly revenue and unit sales for 1,697 unique game titles released for the three consoles during this time period. The data set was collected from 30 of the largest retailers in the United States, retailers that account for around 85% of video game sales. The NPD Group extrapolated the set for the entire U.S. market. I take the ratio of revenue over unit sales in every month to calculate the game price. Additionally, I collected the data on user and critic rating scores for each game title from several large websites such as IGN, GameRankings, GameSpot, and Gamasutra.

The top rows of Table 2 provide the basic information of the three consoles. Up to February 2002, the installed bases of console users in the U.S. market for the Saturn, PS, and N64 were 1.28 million, 28.25 million, and 17.17 million, respectively. Even though the Saturn was the first mover, the console became the "other" system barely two years after its release, running a distant third behind its two rivals. The middle rows of Table 2 display the basic statistics of all video games released for the three consoles. In total, there were 240 Saturn game titles, 1,172 PS game titles, and 385 N64 game titles. The total game sales were around 8.09 million, 300.02 million, and 111.55 million for the three platforms, respectively. PS games took 71% of the market share, while Saturn games' market share was squeezed to below 2%. The average units sold (revenue) per title for the games released on the Saturn, PS, and N64 were 0.03 million (\$1.25 million), 0.26 million (\$8.47 million), and 0.39 million (\$18.73 million), respectively. On average, N64 games generated the highest sales, while Saturn games generated the lowest sales. I use only the sample of sports games to estimate the game market for two reasons: first, it is relatively easy to sort sports games into groups; and second, using a smaller sample reduces the estimation time. The data used in the estimation contain 397 sports games divided into 29 submarkets. The bottom rows of Table 2 provide basic statistics of sports games. I have also checked other patterns of sports games. All the evidence shows that the sample of sports games is representative of the whole sample.

4.1.1. Positive Indirect Network Effects. Buyers adopt a console to access its video games, and game sellers make their games compatible with a console to reach the console users. The installed base of console users is therefore largely contingent on the current

Table 2. Statistics of the U.S. Fifth-Generation Video Game Industry

	Sega Saturn	PlayStation	Nintendo 64
Release date	May 1995	Sept. 1995	Sept. 1996
Provider	Sega	Sony	Nintendo
CPU bits	32	32	64
Microprocessor clock speed (MHz)	28	33.87	93.75
Starting price (\$)	399.9	299.7	199.8
Ending price (\$)	41.0	112.2	87.1
Average unit sales per month (million)	0.02	0.36	0.26
Installed base (million)	1.28	28.25	17.17
	All games		
Number of active titles	240	1172	385
Total unit sold (million)	8.09	300.20	111.55
Units sold per title (million)	0.03 (0.04)	0.26 (0.48)	0.39 (0.67)
Revenue per title (million)	1.25 (1.61)	8.47 (26.71)	18.73 (34.69)
Life span (months)	41 (15)	36 (19)	34 (13)
	Sports games		
Number of active titles	53	263	81
Total unit sold (million)	1.55	71.75	24.39
Units sold per title (million)	0.04 (0.04)	0.29 (0.42)	0.34 (0.37)
Revenue per title (million)	1.49 (1.77)	9.99 (14.88)	16.57 (19.74)
Life span (months)	38 (15)	36 (19)	37 (19)

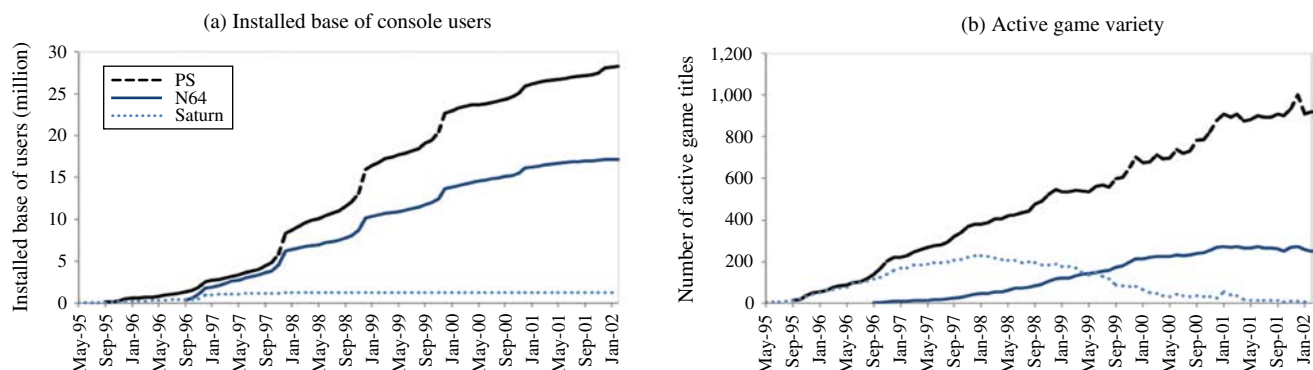
Notes. Summary statistics for the Saturn are for the 82-month period from May 1995 to February 2002, statistics for the PS are for the 78-month period from September 1995 to February 2002, and statistics for the N64 are for the 66-month period from September 1996 to February 2002. Ending price, installed base, total active titles, and total units sold with any console are for February 2002, the last month in the sample. Numbers in parentheses are standard deviations.

and expected availability, quality, and price of affiliated games, and the variety of games affiliated with a console depends on how many users have adopted and are expected to adopt that console. Figure 1(a) presents each platform's installed base of console users, and Figure 1(b) presents the number of active game titles affiliated with each console during the sample period. Overall, Figure 1 shows that the installed base of console users and the active game variety have the same growth pattern, implying a positive correlation between buyers' entry and game sellers' entry.

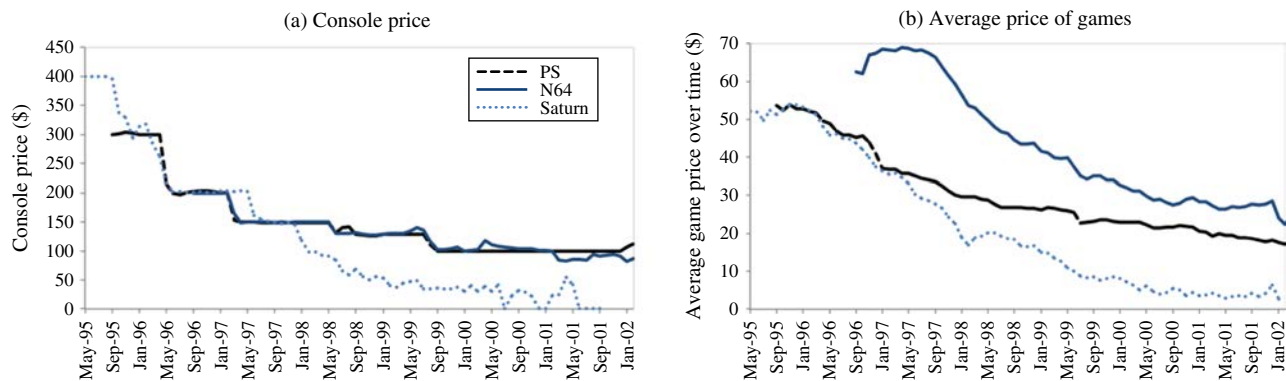
4.1.2. Console Price and Game Price Decline Over Time. Console prices are shown in Figure 2(a). Overall, console prices were declining over time. It is widely

speculated that all the major consoles were initially sold at a price near marginal cost. Industry analysts and many academic articles (see Liu 2010) also report that console prices fell slower than production costs, and thus the margin actually increased over time. Figure 2(b) shows the average price of games affiliated with each console over time. Initially, N64 games were much more expensive than others, but the price gap became smaller over time. Game prices increased slightly during the first few months after a console was introduced and thereafter declined smoothly over time, probably because most games available in early time periods were provided by console makers who strategically charged lower prices than third-party game sellers, and

Figure 1. (Color online) Installed Base of Console Users and Active Game Variety



Notes. (a) The installed base of console users is measured by the cumulative units sold of each console in millions. The monthly sales of the Saturn were below 0.5 million units after January 1997. (b) Active game variety refers to the number of game titles with positive sales.

Figure 2. (Color online) Console Price and Average Game Price Over Time

Notes. (a) Average monthly (nominal) prices faced by buyers in retailer stores for each console in the U.S. market from May 1995 to February 2002. (b) Average monthly (nominal) prices of video games released for each console.

the share of in-house games declined as more third-party games came to the market over time.

4.1.3. Game Price and Sales Decline with Age. In Figure 3, the horizontal axis is the game age measured by the months since introduction, and the vertical axis is the average game price in (a) and the average unit sales in (b). The average game price was around \$45 per copy at release and then dropped to about \$23 the following year. The average game sales were around 40 thousand units in the first month and then fell to around five thousand units per month after the first year. What drives game prices and sales to drop so quickly? A falling-cost explanation is not convincing for this industry. Once a video game is developed, game sellers only need to pay royalty fees to the console maker and their own production costs. Both costs remain roughly constant per unit over time.⁸ As explained in the introduction (also see Nair 2007, Hendel and Nevo 2013), the most reasonable explanation is sellers' intertemporal price discrimination. Additionally, the entry of new games leads to more intense competition and thus induces the manufacturers of existing game titles to cut their prices.

4.2. Econometric Details

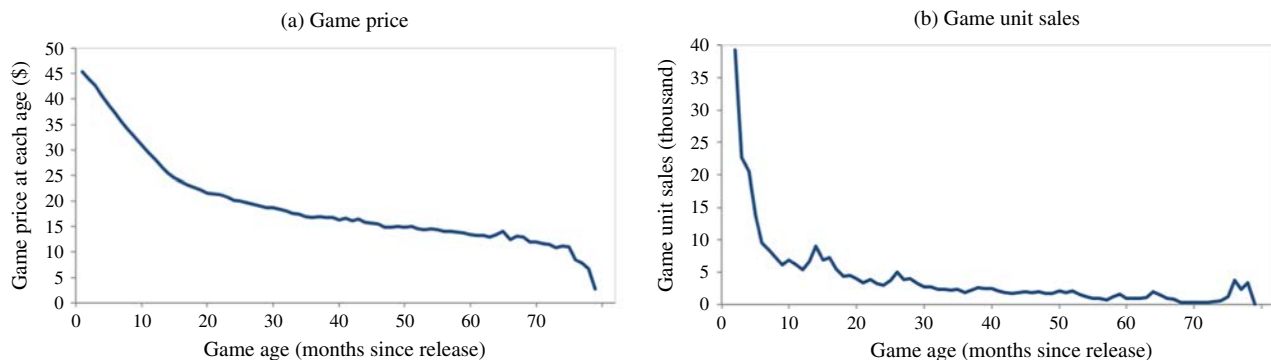
4.2.1. Buyer Heterogeneity. In this model, the existence of buyer heterogeneity generates the incentive for

sellers to intertemporally price discriminate. To model buyer heterogeneity, I adopt Nair's (2007) random coefficients approach with discrete approximation. Formally, the distribution of individual-specific parameters in platform utility function, γ_i , and individual-specific parameters in affiliated product utility function, φ_i , is assumed to be

$$(\gamma_i, \varphi_i) = \begin{cases} (\gamma_1, \varphi_1) & \text{if } i = 1, \\ (\gamma_1 + D_i^\gamma, \varphi_1 + D_i^\varphi) & \text{if } i = 2, \dots, I, \end{cases} \quad (19)$$

where the first type is treated as a "base" type and the parameters of the other types are expressed as deviations from the base type. For simplicity but without loss of generality, I assume two buyer types, high-type buyers and low-type buyers.⁹ At the outset, it is necessary to choose an initial number of buyers, N_0 . Once this is pinned down, the future distribution of each buyer type is determined by the buyer purchase decisions of console and games. Since a television is a necessary equipment for playing video games, I assume that the households with televisions are the potential buyers of video game consoles. According to census data, approximately 96 million U.S. households owned a television in 1995. Hence, I set N_0 to be 96 million.

4.2.2. Platform Characteristics. The vector of platform characteristics includes platform price, platform-specific dummy variables, and holiday dummy

Figure 3. (Color online) Game Price and Unit Sale at Each Age

variables. I allow that buyers have heterogeneous sensitivity to platform price and heterogeneous valuations of platforms. Formally, the term of platform characteristics is written as

$$X_{it}\gamma_i = \gamma_i^P P_{it} + \gamma_i^{PS} 1\{l = PS\} + \gamma_i^{N64} 1\{l = N64\} \\ + \gamma_{Nov.} 1\{t \text{ is Nov.}\} + \gamma_{Dec.} 1\{t \text{ is Dec.}\}.$$

4.2.3. Affiliated Product Characteristics. The vector of affiliated product characteristics includes platform-specific dummy variables, online rating score, product age, and holiday dummy variables. Let age_{jt} denote the months after release. To capture the fact that buyers' utility declines with game age, with different declining rates for new games and old games, I treat a game as a new game if it has been in the market less than one year, and as an old game if it has been in the market longer than one year. Formally, the term of affiliated product characteristics is

$$x_{jt}\psi = \psi_{PS} 1\{j \in PS \text{ games}\} + \psi_{N64} 1\{j \in N64 \text{ games}\} \\ + \psi_{rating} Rating_j + \psi_{new} \min(age_{jt}, 12) \\ + \psi_{old} \max(age_{jt} - 12, 0) + \psi_{Nov.} 1\{t \text{ is Nov.}\} \\ + \psi_{Dec.} 1\{t \text{ is Dec.}\}.$$

4.2.4. Discount Factors. Previous studies have pointed out that the discount factor of dynamic models is usually not identified, so I do not attempt to estimate it. Instead, I set all discount rates (i.e., β_1 , β_2 , β_3 , and β_4) to be 0.996, which has been typically assumed for monthly data.

4.2.5. Potential Entrants. Like most other available data sets, my data set includes information only on actual entrants, and not on those who consider entering but choose not to. It is difficult to define who are the potential entrants in the video game market because a significant proportion of video games are developed by small studios. Following Seim (2006), I assume a fixed number of potential entrants for each time period (month). I use a potential entrant pool of five firms, which for most submarkets is more than twice the number of actual entrants. This assumption affects the magnitude of the entry cost for each console. However, it does not affect the relative cost of one console compared to the cost of a particular console. This paper focuses on why one console has few games but another console has many games. Hence, the relative cost is more important than the absolute cost. This idea is the same as in Lee (2013), who normalizes the porting cost for the PS2 to be zero and estimates the additional entry cost of other consoles compared to the PS2. The ratings of potential entrants are drawn from the empirical distribution of the realized ratings, which is assumed to follow an i.i.d. normal distribution.¹⁰

4.2.6. Prior Distribution. To estimate the model, it is necessary to specify the prior distribution for the parameters to be estimated. All coefficients in the utility functions are assumed to follow normal distributions with means of zero and large standard deviations. The initial share of high-type buyers (δ) follows a uniform distribution on the interval $[0, 1]$. To guarantee that cost parameters and standard deviations are non-negative, their prior distributions are assumed to be log-normal with means of zero and large standard deviations.

4.2.7. Initial Guess of Equilibrium Strategies and Value Functions. To compute the initial guess for buyers' value functions and incumbent sellers' value functions, I assume that both buyers and game sellers are myopic. The initial guesses for the prices of affiliated products are the predicted values from a hedonic regression of price on state variables. I compute the initial guess of the entry probability based on the initial value functions.

4.3. Posterior Statistics

I draw 100,000 samples from the posterior distribution and use the last 50,000 samples to derive the posterior means and standard deviations. I also compute the posterior means and standard deviations from the last 25,000 samples. Table 3 reports the posterior statistics of the two samples. The findings in the two sets of samples are not statistically different. I repeat this procedure several times and find that the posterior statistics are robust to initial values of parameters that are drawn from their prior distributions. I conclude that the samples I use to compute the posterior statistics are drawn from a stable distribution.

The estimates of parameters in the platform utility equation are consistent with our expectation. The N64 fixed effects are 0.378 and 0.104 for the high-type buyers and low-type buyers, respectively. However, the PS fixed effects are much lower, only 0.036 and 0.013 for the two types. This indicates that the PS's quality is not significantly higher than the Saturn's. Hence, the Saturn's failure in the war with the PS was not because its console's quality was inferior to that of the PS. High-type and low-type buyers' price sensitivities are -0.015 and -0.062 , respectively. The weight of affiliated products in a platform's value is 0.457. This number is not very high, probably because I only use the sports games in the estimation. Buyers obtain higher utility from purchasing consoles during the holiday season. High-type buyers correspond to 18.9% of the potential market at the beginning of the console life cycle. The relatively small share of high-type buyers partly explains why the prices decline so quickly.

The estimates of parameters in the game utility equation are consistent with our expectation. Nintendo 64

Table 3. Posterior Means and Standard Deviations

	Last 50,000 samples		Last 25,000 samples	
	Mean	Std. dev.	Mean	Std. dev.
Block 1: Platform adoption				
α (importance of affiliated products)	0.457	0.120	0.454	0.119
γ_1^P (H-type buyers' price sensitivity)	−0.015	0.006	−0.015	0.006
γ_1^{PS} (H-type buyers' PS FE)	0.036	0.021	0.035	0.020
γ_1^{N64} (H-type buyers' N64 FE)	0.378	0.161	0.373	0.152
γ_2^P (L-type buyers' price sensitivity)	−0.062	0.010	−0.062	0.009
γ_2^{PS} (L-type buyers' PS FE)	0.013	0.006	0.013	0.006
γ_2^{N64} (L-type buyers' N64 FE)	0.104	0.052	0.104	0.052
$\gamma_{Nov.}$ (Nov. FE)	1.012	0.278	1.013	0.259
$\gamma_{Dec.}$ (Dec. FE)	2.371	0.491	2.368	0.490
σ_ϵ (std. dev. of platform utility shocks)	0.347	0.126	0.349	0.127
δ (initial share of H-type buyers)	0.189	0.088	0.189	0.088
Block 2: Affiliated products				
φ_1 (H-type buyers' price sensitivity)	−0.022	0.003	−0.022	0.003
φ_2 (L-type buyers' price sensitivity)	−0.060	0.030	−0.060	0.028
ψ_{PS} (PS games FE)	0.100	0.031	0.100	0.031
ψ_{N64} (N64 games FE)	1.573	0.533	1.568	0.519
ψ_{rating} (online rating score)	0.172	0.110	0.175	0.109
ψ_{new} (age for new games)	−0.383	0.082	−0.389	0.071
ψ_{old} (age for old games)	−0.155	0.069	−0.157	0.068
$\psi_{Nov.}$ (Nov. FE)	0.278	0.024	0.278	0.024
$\psi_{Dec.}$ (Dec. FE)	0.711	0.082	0.711	0.082
c_{Saturn} (unit cost of Saturn games)	11.213	2.264	11.223	2.199
c_{PS} (unit cost of PS games)	10.589	1.671	10.583	1.555
c_{N64} (unit cost of N64 games)	18.874	1.662	18.874	1.595
σ_ξ (std. dev. of demand shocks)	2.031	0.426	2.078	0.422
σ_ζ (std. dev. of supply shocks)	0.988	0.326	0.989	0.327
$\sigma_{\xi\zeta}$ (correlation)	0.322	0.041	0.328	0.038
Block 3: Entry of affiliated product sellers				
λ_{Saturn} (mean of entry cost to Saturn games)	4.243	0.555	4.244	0.555
λ_{PS} (mean of entry cost to PS games)	3.268	0.678	3.265	0.674
λ_{N64} (mean of entry cost to N64 games)	4.590	0.366	4.905	0.362
σ_v (std. dev. of game entry cost shocks)	2.601	0.216	2.603	0.216

Note. H, high; L, low; FE, fixed effects.

games generate the highest utility because the console is more technologically advanced than the other two. PS games are not significantly better than Saturn games, which suggests that the Saturn's failure in the war with the PS was not because its games' quality was inferior to that of PS games. Buyers favor the games with high online ratings and obtain higher utility from purchasing games during the holiday season. They favor games that have been in the market for a short time, partly because most sports games are designed based on the latest tournaments. High-type buyers' price sensitivity to software is −0.022, and low-type buyers' price sensitivity to affiliated games is −0.060.

The unit costs are \$11.2, \$10.6, and \$18.9 for Saturn games, PS Games, and N64 games, respectively. As Coughlan (2001) reported, the production and packaging cost for a 32-bit CD-ROM game is around \$1.5 per disc. Therefore, the royalty rates charged by Sega and Sony were around \$9.7 and \$9.1 per copy sold, respectively. The unit cost of N64 games is much higher

than Saturn games and PS games because Nintendo used ROM cartridges to store games, so their production expense was much higher than the production expense for the compact disc format that competitors used. The estimates of royalty rates are close to the numbers reported by Coughlan (2001) and many academic articles on the video game industry such as Nair (2007), Liu (2010) and Dube et al. (2010).

The average entry costs of Saturn games, PS games, and N64 games are \$4.2 million, \$3.3 million, and \$4.6 million, respectively. The standard deviation of entry cost is \$2.6 million. Saturn games' research and development cost was on average significantly higher than that of PS games, partly due to the Saturn's dual-CPU architecture and more complex graphics.

4.4. Counterfactual Experiments

In this section, I treat the posterior means of the last 50,000 samples as the estimated values of the parameters. I first numerically solve for the equilibrium of

the proposed dynamic model using the nonparametric approximation and interpolation method described in Section 3. The equilibrium values are the predictions of the benchmark model that treat both buyers and sellers as forward-looking agents. The black line of Figure 4 shows the weighted average price of affiliated products in age, and the black lines of Figure 4 show the number of active buyers and number of active sellers for each console over time. After that, I conduct counterfactual experiments to examine the implications of platform participants' dynamic behavior and to explore the possible causes of Sega's failure.

4.4.1. Dynamic Behavior of Platform Participants. The first two experiments examine the impact of buyers' and sellers' dynamic behavior on the prices of affiliated products, and the third and fourth experiments examine the impact of their dynamic behavior on the installed bases of platform participants.

Experiment 1: Buyers' dynamic behavior in purchasing affiliated products. In the first experiment, buyers who have already adopted a platform are myopic in the affiliated products market; in other words, they do not realize that they can make a purchase decision in a later period. I numerically solve the model by setting buyers' discount factor in the affiliated product market to be zero, that is, $\beta_2 = 0$. To isolate the impact of buyers' dynamic behavior on the dynamic pricing, the new entry of buyers to the affiliated product market (i.e., buyers' adoption of platforms) and the new entry of sellers are kept the same as predicted by the benchmark model. The purple line in Figure 4 represents the average prices in this counterfactual scenario. It shows that the prices of affiliated products with myopic buyers are

higher than those with forward-looking buyers. This is because myopic buyers do not trade off between enjoying a higher consumption value today and waiting for lower prices so that sellers do not need to "compete" with future selves. Moreover, the price gap narrows as a product becomes older because buyers' value of waiting declines over time.

Experiment 2: Incumbent sellers' dynamic behavior in setting prices. The second experiment is similar to the first one except that incumbent sellers of affiliated products are myopic, that is, $\beta_3 = 0$. Again, the new entry of buyers and sellers to the affiliated product market are kept the same as predicted by the benchmark model. The orange line in Figure 4 gives the weighted average price of affiliated products in this counterfactual scenario. It indicates that myopic sellers would charge lower prices than forward-looking sellers. The reason is that myopic sellers do not account for that a higher price today would increase the potential buyers in the future and thus improve the profitability of later periods, nor account for that a higher price today would reduce buyers' value of waiting in earlier periods and thus improve the profitability of earlier periods.

Experiment 3: Buyers' dynamic behavior in adopting platforms. In this counterfactual experiment, buyers who have not adopted any platform are myopic when making their platform adoption decisions, that is, $\beta_1 = 0$. Intuition suggests that buyers' dynamic behavior has two opposite effects on the platform adoption. On the one hand, the consumption value of a platform is substantially lower for myopic buyers who do not account for the consumption values of that platform in later periods. This difference shrinks as the products approach the end of the product life cycle. As a result, myopic buyers are much less likely to adopt platforms than forward-looking buyers if everything else is equal. Because of the positive indirect network effects, fewer sellers would join, and in turn, fewer buyers would adopt platforms. On the other hand, when myopic buyers make their platform adoption decisions, they do not realize that they can adopt them in a later period. This implies that the value of the no-purchase option for myopic buyers is lower than that for forward-looking buyers, and this difference is bigger in the early stage of the product life cycle. Hence, myopic buyers are more likely to adopt platforms than forward-looking buyers, if everything else is equal. The purple lines in Figure 5 represent simulation outcomes of this counterfactual scenario. The results show that both the installed base of buyers and number of active sellers are substantially lower than the counterparts when buyers are forward looking. This suggests that the first effect dominates the second one.

Figure 4. (Color online) Impact of Dynamic Behavior on Prices of Affiliated Products

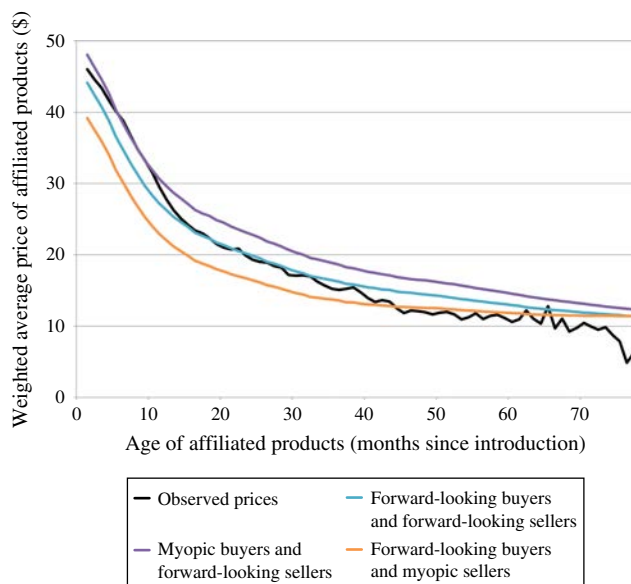
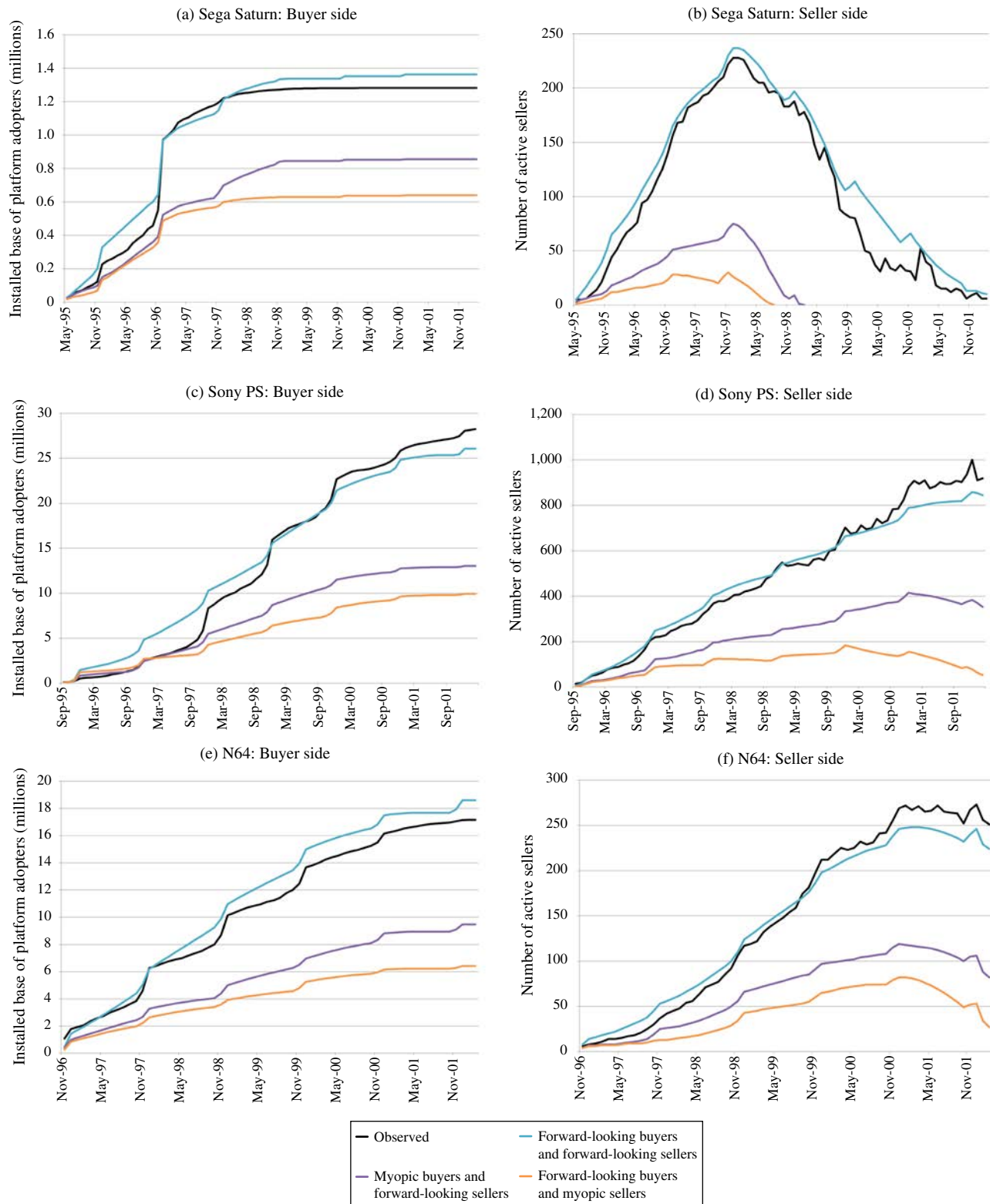


Figure 5. (Color online) Impact of Dynamic Behavior on Installed Bases of Platform Participants

Experiment 4: Sellers' dynamic behavior in joining platforms. In this counterfactual experiment, sellers who have not entered are myopic when making their entry decisions, that is, $\beta_4 = 0$. The orange lines in Figure 5 show that the installed base of platform adopters

and the number of active buyers are substantially lower than the counterparts when sellers are forward-looking agents. This is because myopic sellers do not account for the future profits when they make the entry decisions so they are much less likely to join a plat-

Table 4. Simulation Results of Sega's Various Subsidy Packages

		Experiment 5 (\$)			Experiment 6 (\$ m)			Experiment 7 (\$)	
	Observed	−0	−50	−100	−0	−0.5	−1	−0	−5
Sega Saturn:									
Installed base of buyers (million)	1.28	6.68	5.04	3.98	5.13	4.22	3.07	3.14	2.01
Total sales of sports games (million)	2.11	14.24	10.46	7.27	11.84	10.79	8.53	8.23	5.59
Total subsidy (million dollars)	—	297.40	248.57	180	68	65	61		
Increased royalty revenue from sports games (million dollars)	—	117.66	81.00	50.05	94.38	84.20	62.27	38.68	26.27
Total net gain/loss (million dollars)	—	164.29	32.87	−38.68	119.46	86.93	5.77	82.48	26.29
Sony PlayStation:									
Installed base of buyers (million)	28.25	26.12	32.67	38.24	27.45	30.94	32.21	27.86	32.77
Total sales of sports games (million)	75.97	70.08	85.89	91.96	73.92	83.11	87.94	74.42	95.82
Total net gain/loss (million dollars)	—	−241.48	−137.80	−109.24	−84.05	−500.31	−505.88	−64.57	−1,342.55
Nintendo 64:									
Installed base of buyers (million)	17.17	16.95	14.88	12.15	17.08	15.87	12.87	17.05	16.46
Total sales of sports games (million)	27.84	26.37	23.79	22.56	27.16	25.06	22.91	27.11	25.99

Note. All numbers are for February 2002, the last month of the sample.

form. As a result, buyers are much less likely to adopt a platform due to less variety of affiliated products available for them to purchase and consume if they adopt a platform. Because of the positive indirect network effects, fewer buyers would adopt platforms, and in turn, fewer sellers would join.

4.4.2. Sega Saturn's Failure. It has been speculated that Sega was shaken out the market because it did not successfully bring both sides on board during the launch stage. Experiments 5–7 explore whether Sega would have survived if it had subsidized buyers' entry or game sellers' entry or lowered the transaction fee. Table 4 reports the three sets of results. In each counterfactual scenario, I lower Sega's one price and simulate buyers' and game sellers' responses. As stressed in Section 2, I do not model the price competition among platform providers. Incorporating competing platforms' equilibrium responses requires a dynamic oligopolistic equilibrium model for both consoles and affiliated products, which is too complicated. Instead, I consider different reactions from competing platforms and simulate the installed base of platform participants under each of these possible reactions. Even though the counterfactuals are "partial," the simulation results still shed some light on how a platform's prices affect that platform's network size. The simulation quantitatively measures how responsive buyers and game sellers are to a platform's prices. The results also highlight the importance of getting both sides on board during the launch stage.

Experiment 5: Sega subsidizing buyers' entry. Figure 2(a) shows that Sega's console prices were \$100 higher than the PS's prices at the same age for the first two years. A high console price discouraged buyers' entry and hence game sellers' entry. It is speculated that this contributed to Sega's failure. This counterfactual

exercise examines what would have happened if Sega had lowered the console price by \$100 during the first two years. I consider three different reactions of Sony: maintaining its console price, cutting its console price by \$50, and fully matching Sega's price cut. If Sony had maintained its console prices, the number of Sega console users would have increased from 1.28 to 6.68 million units at the end of the sample period (February 2002), and its total sales of sports game would have increased from 2.11 to 14.24 million units. Because the \$100 subsidy is not a negligible number, the impacts on console adoption and game sales are considerably large. Because of the lower console price, Sega would have needed to subsidize 297.40 million dollars on the buyer side of the market. However, the increased sales of the sports games would have led to an extra 117.66 million dollars in royalty revenues collected from game sellers. Since I only know the increased royalty revenue from sports games, the total net gain from this subsidy package is unknown. If nonsports games behave in the same way as sports games, then the increased royalty revenue from all games would be scaled up to 532.80 million dollars. As a result, the total net gain would be 164.29 million dollars. Therefore, reducing the console price during the launch stage would have been very effective in promoting Sega's platform size and increasing its profit. Sony would have suffered from Sega's lower console price, because the PS's technology and entry time were almost the same as the Saturn's, and hence the PS was the major rival of the Saturn. However, the N64 would not have been affected as much, because the N64 was more technologically advanced and entered the market one and a half years later than the Saturn and hence was not the Saturn's direct competitor. Even if Sony matched Sega's price cut, Sega's installed base of users would still be 3.98 million units, and its total sales of sports games

would be 7.27 million units. The results of this experiment suggest that high entry cost to buyers was one of the main causes of Sega's failure. This is because, in two-sided markets, the responsiveness of the quantity demanded not only depends on buyers' price elasticity on the good, but also on the strength of the indirect network effects. A lower console price attracts new buyers and hence new game sellers, new games attract extra new buyers, and so on. If buyers are very sensitive to console price and game sellers highly value the installed base of buyers, a change in the console price would have a large impact.

Experiment 6: Sega subsidizing affiliated product sellers' entry. Table 3 indicates that Saturn's entry cost for game sellers is on average much higher than that of the PS. High entry cost discourages game sellers' entry and thus discourages buyers' entry. If game sellers are very sensitive to entry cost and users highly value the game variety, subsidizing game sellers' entry would have a considerable impact. Experiment 6 simulates the installed base of platform users if Sega had given game sellers a subsidy of one million dollars per game title.¹¹ The results show that Sega would have gained 119.46 million dollars if Sony did not subsidize any dollar, 86.93 million dollars if Sony matched half amount of Sega's subsidy, and 5.77 million dollars if Sony fully matched Sega's subsidy package. This suggests that subsidizing game sellers' entry would also have improved Sega's performance.

Experiment 7: Sega lowering the royalty rate. The estimates of the unit costs of Sega games is just a little higher than that of PS games. However, it is still interesting to see what would happen if Sega had reduced its royalty rate. Lower royalty rate would have led to a lower game price. This would have increased the attraction of the Saturn and hence encouraged buyer entry. The positive indirect network effect between buyers' entry and game sellers' entry would have enlarged this impact. In this counterfactual, I simulate the outcomes if Sega had aggressively reduced its royalty rate by five dollars per unit. With the proposed new royalty rate, Sega would have gained 82.48 million dollars if Sony maintained its royalty rate and 26.29 million dollars if Sony matched Sega's cut.

5. Conclusions

This paper developed a model of two-sided markets, with a focus on the dynamic strategic interactions between platform participants. On the buyer side, the value of a platform depends on the expected value of optimally purchasing the products associated with that platform. Moreover, the potential buyers for an affiliated product depend on buyers' platform adoption. The demand system of the model accounts for dynamic

selection of forward-looking and heterogeneous buyers into platforms for affiliated products and allows for the contingency of platforms' value on the availability and prices of the affiliated products. By incorporating the complementarity between platforms and affiliated products, I was able to examine the effects of subsidizing sellers' entry on buyers' adoption of platforms. On the seller side, sellers compete within a submarket. Incumbents choose their prices, and potential entrants choose whether to enter and strategically account for competitors' reactions and buyers' responses. I have investigated the impacts of subsidizing buyers' entry on sellers' entry and price choices.

This paper also presented a practical Bayesian MCMC procedure for structural estimation of dynamic models. The estimation procedure combines approximation and interpolation methods. The computational burden is relatively light. This procedure can be applied to estimate other dynamic models, especially those with unobserved heterogeneity and large state space.

This analysis framework can be applied to various two-sided markets with durable platform intermediaries and affiliated products. I have implemented it to the fifth generation of the U.S. video game market. I have examined the implications of platform participants' dynamic behavior on platform adoption and the affiliated product market. I have also examined Sega's alternative strategic options, such as subsidizing buyers' entry, subsidizing game sellers' entry, or reducing the royalty rate. For each alternative option, I simulated the installed base of platform participants and Sega's potential gain. The results show that Sega would have avoided being shaken out of the market if it had priced the two sides properly.

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Endnotes

¹Theoretical studies include Coase (1972), Stokey (1982), and many subsequent papers, and empirical studies include Nair (2007), Gowrisankaran and Rysman (2012), and Conlon (2012).

²Ruling out buyer multihoming may potentially cause biases. This paper does not allow for buyer multihoming for two main reasons. First, allowing for buyers to multihome would significantly complicate the computation because buyers' inventory would become a state variable. Second, precise data on the degree of buyer multihoming is rarely available for researchers.

³Here, I implicitly assume that only platform adopters are the potential buyers of the affiliated products. In reality, people may buy the affiliated products as gifts, which would potentially bias the estimates. Unfortunately, researchers seldom know whether people buy a product for self-consumption or for a gift. If the share of gift purchases is small, the potential bias should be negligible.

⁴See Coughlan (2001), Nair (2007), Liu (2010), and Dube et al. (2010).

⁵Notice that this assumption on entry cost shocks may not hold if we consider learning by doing or technology spillover effect.

⁶See McCulloch and Rossi (1994), Albert and Chib (1993), and Imai et al. (2009).

⁷Take two types for example. Let δ_i be the share of high-type buyers at time t , and let $b_{jt}^i(p_i)$ be the i -type buyer's probability of purchasing product j at time t . Then, the market share of product j at time t can be written as $s_{jt} = \delta_i(1 - b_{jt}^H(p_i)) + (1 - \delta_i)(1 - b_{jt}^L(p_i))$. First, consider one submarket at a specific time period. Since the share of high-type buyers is the same across products within a submarket, the sales variation across products to the price changes can help to identify the price coefficients. Second, the sales variations as the price changes across submarkets can help to identify the buyer heterogeneity. Third, consider one submarket but two different time periods (say, $t = 1, 2$). Because of buyer heterogeneity, high-type buyers leave the market earlier than low-type buyers, and hence $\delta_1 > \delta_2$. Then, the change of the market shares of the same product over time can help to identify the share of high-type buyers. Additionally, equilibrium prices are lower in the second period, and thus the extent to which $s_{j1} > s_{j2}$ tells us more about the extent to which there is heterogeneity.

⁸Coughlan (2001) reports that production and packaging costs for 32-bit CD-ROM games remain roughly constant at \$1.50 per disc, and Nair (2007) reports that the royalty fee for the Sony PlayStation-compatible games was preannounced and held fixed at \$10 by Sony throughout the life cycle.

⁹I began by allowing for two types and then increased the number to be three, four, and five. I found that the changes were negligible in both model fit and predictions. Hence, I only report the results for the two-type case.

¹⁰This tends to overestimate the entry cost, because products that actually do enter the market are more likely of higher quality than those products that do not enter.

¹¹Even though a game seller's entry cost does not directly go to platform providers' pockets, platforms can still strategically influence these game sellers' entry fees, either by offering easier-learning development technology or by directly subsidizing game research and development.

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